

JOBA, Oman

WMO: 412870

Lat: 20.8681N

Lon: 58.2475E

Elev: 112

StdP: 14.64

Time Zone: 4.00 (E04)

Period: 05-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	54.8	57.2	25.0	19.2	81.3	29.4	23.5	82.2	24.7	70.6	21.3	73.6	5.8	350	0.469

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
6	21.2	108.7	71.2	105.4	72.1	102.6	73.2	81.3	92.5	80.6	90.8	80.0	90.1	13.9	270

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
79.2	152.1	84.4	78.4	148.2	84.4	77.6	144.2	84.4	45.2	92.3	44.5	91.1	43.8	90.1	89.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 27.4	25.0	22.8	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(5)			39.5	81.3	4.7	1.5	36.2	82.3	33.5	83.2	30.8	84.0	27.5	85.1	(4)	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	81.7	69.8	72.6	78.9	85.7	91.3	91.4	86.9	84.9	84.7	83.9	78.3	72.0
	DBStd	7.71	2.93	3.72	3.65	3.66	2.96	3.88	3.43	2.75	2.50	2.63	2.99	3.15
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD65	4	3	1	0	0	0	0	0	0	0	0	0	0
	CDD50	11588	614	634	896	1072	1281	1241	1144	1081	1042	1052	848	683
	CDD65	6116	151	215	431	622	816	791	679	616	592	587	398	218
	CDH74	71187	1210	1865	4473	7680	11487	11146	8031	6525	6353	6672	3940	1805
CDH80	37381	200	580	2157	4383	7148	6895	4104	3056	3063	3575	1757	463	
Wind	WSAvg	10.4	7.6	8.5	9.2	11.5	12.8	12.9	14.4	13.6	11.7	8.2	6.7	7.6
Precipitation	PrecAvg	2.2	0.1	0.1	0.2	0.2	0.0	0.6	0.2	0.0	0.0	0.2	0.5	0.1
	PrecMax	7.6	2.3	0.8	2.5	3.0	0.0	5.4	2.4	0.5	0.0	1.7	3.9	0.6
	PrecMin	0.3	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
	PrecStd	1.9	0.4	0.2	0.6	0.6	0.0	1.3	0.6	0.1	0.0	0.4	0.9	0.2
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	87.2	92.6	100.7	106.5	112.1	115.0	109.1	102.4	103.1	103.1	96.3	89.5
		MCWB	62.8	64.5	64.5	67.5	68.6	69.9	76.3	77.5	73.8	67.9	67.2	65.0
	2%	DB	84.1	88.6	96.6	103.8	108.1	109.8	102.4	99.0	98.8	100.0	93.4	86.6
		MCWB	63.6	64.7	64.9	67.2	70.4	71.4	77.4	77.2	74.4	68.6	68.4	65.3
	5%	DB	82.1	86.0	93.3	100.8	104.9	106.1	98.6	95.7	96.0	97.5	91.1	84.6
		MCWB	64.0	65.3	65.9	68.1	72.3	74.4	78.1	76.8	74.3	69.6	68.7	65.3
	10%	DB	80.5	83.5	90.1	97.4	101.9	101.7	95.2	92.9	93.0	94.7	88.7	82.6
		MCWB	63.3	65.3	66.3	68.7	74.5	76.0	77.7	76.7	75.1	69.7	69.2	65.1
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	70.6	72.9	76.5	79.5	82.2	82.5	81.2	80.0	78.8	78.1	76.5	72.9
		MCDB	78.9	80.7	84.0	87.6	94.5	95.6	96.5	95.2	92.9	87.2	83.7	80.3
	2%	WB	68.4	71.0	75.3	78.1	81.0	81.5	80.1	78.7	77.8	77.1	75.3	70.6
		MCDB	77.6	79.3	82.3	85.3	90.6	93.0	94.5	93.8	89.8	85.7	83.7	79.9
	5%	WB	67.3	69.8	73.8	77.1	80.3	80.9	79.2	77.9	77.2	76.3	73.8	69.1
		MCDB	77.0	78.6	81.7	84.3	89.0	91.1	92.5	91.4	88.0	84.7	82.8	78.8
	10%	WB	66.1	68.8	72.7	76.1	79.6	80.3	78.4	77.1	76.5	75.5	72.4	67.7
		MCDB	75.7	77.7	81.1	84.0	88.4	90.0	90.6	89.3	86.5	83.8	82.0	77.7
Mean Daily Temperature Range	5% DB	MDBR	23.5	23.9	24.8	24.3	23.2	21.2	18.3	17.8	19.4	23.5	22.5	23.1
		MCDBR	24.6	26.2	29.1	28.5	27.3	28.6	23.8	22.4	23.6	27.7	25.1	24.8
		MCWBR	10.0	9.4	9.7	10.0	8.2	8.0	4.6	4.4	4.4	7.7	8.7	9.8
	5% WB	MCDBR	22.9	22.1	20.5	22.2	21.4	18.9	19.2	19.0	18.9	20.8	21.8	21.3
		MCWBR	10.3	8.0	6.5	7.1	5.1	4.2	4.4	4.8	4.0	5.3	8.2	9.3
Clear-Sky Solar Irradiance	taub	0.413	0.454	0.507	0.570	0.639	0.756	0.828	0.776	0.623	0.469	0.438	0.416	
	taud	2.133	1.997	1.852	1.741	1.628	1.459	1.386	1.452	1.677	1.986	2.084	2.121	
	Ebn at Noon	269	265	256	242	223	197	183	194	223	257	259	263	
	Edh at Noon	45	55	66	75	83	97	105	99	78	55	47	44	
All-Sky Solar Radiation	RadAvg	1564	1809	2084	2292	2381	2244	2062	2058	2061	1904	1617	1473	
	RadStd	40	47	72	59	31	120	120	65	48	40	42	52	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (1 neighbor)	N/A	N/A	N/A	N/A	-1.13	-1.78	N/A	N/A	N/A	N/A

Nomenclature: See separate page